

Pharmacy V13

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September 8, 2026

Abstract

Study context and scope. Pharmacy V13 is developed in the context of pharmacy operations where purchasing, sales, inventory, suppliers, expiry, and financial transactions generate substantial operational information but do not, by themselves, provide integrated management intelligence. The study addresses the need to transform these routine operational records into structured, interpretable, and evidence-traceable decision support. The scope encompasses the design and integration of an operational pharmacy information system with analytical and management-intelligence capabilities, with particular emphasis on supply chain analysis, inventory management, demand understanding, risk assessment, and model evaluation. The system is intended to support management rather than to replace professional judgement or authorise automated decisions.

Pharmacy V13 is therefore conceived as an integrated management intelligence system that transforms routine pharmacy operations into structured, evidence-based decision support. Rather than functioning solely as a system for recording medicines, purchases, sales, inventory, suppliers, and financial transactions, V13 establishes a progressive analytical architecture linking operational evidence to information, analytical understanding, management intelligence, and accountable decision-making.

The system incorporates a dedicated Supply Chain Analysis and Intelligence Layer, supported by the REEM intelligence engine, to integrate observed purchasing, sales, inventory, supplier, expiry, and demand evidence. Detailed computational outputs from analytical models are transformed into management summaries, visual analytics, risk and exception indicators, model-readiness assessments, and evidence-traceable management signals. The analytical portfolio includes demand forecasting, stock coverage, ABC analysis,

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expiry risk, reorder point, safety stock, economic order quantity (EOQ), and Monte Carlo uncertainty analysis.

A central principle of the architecture is the distinction between observed evidence, derived indicators, model outputs, and decision intelligence. Model execution alone is therefore not considered sufficient for operational use. Models are assessed according to their analytical purpose and evidence requirements. Continuous forecasting models are evaluated through forecast error and residual analysis, optimisation models through parameter and sensitivity analysis, while appropriate binary classification models may be evaluated using confusion matrices, precision, recall, sensitivity, specificity, F1 score, calibration, and receiver operating characteristic (ROC) analysis with area under the curve (AUROC), where adequate labelled historical outcomes exist.

The architecture adopts a progressive model-readiness framework ranging from *Available* and *Partial* to *Ready for Extension* and *Operationally Validated*. This prevents mathematically completed calculations from being interpreted automatically as validated procurement recommendations. Instead, Pharmacy V13 positions analytical models as transparent instruments for management understanding, risk assessment, and decision support.

The study does not claim that every analytical model is operationally validated at the point of implementation. The maturity of each model depends on the availability, quality, duration, and completeness of historical evidence, together with the availability of appropriate model parameters and labelled outcomes where required. Consequently, exploratory and descriptive analyses may be used where evidence is currently limited, while predictive, classification, simulation, and optimisation capabilities are progressively validated as the evidence base matures.

The overall objective is to establish a scientifically defensible and evidence-traceable pathway from observed pharmacy transactions to reliable historical knowledge, validated analytical models, risk-aware prediction, responsible optimisation, and accountable management action. Within this scope, artificial intelligence, statistical modelling, simulation, and decision-support methods are intended to inform management; they do not replace human judgement, professional responsibility, or organisational governance.